CourseLens

An NLP study assistant for finding answers inside preloaded lecture notes.

KAP

Lecture data

TF-IDF search

Evaluation

Next steps

Project Update 2

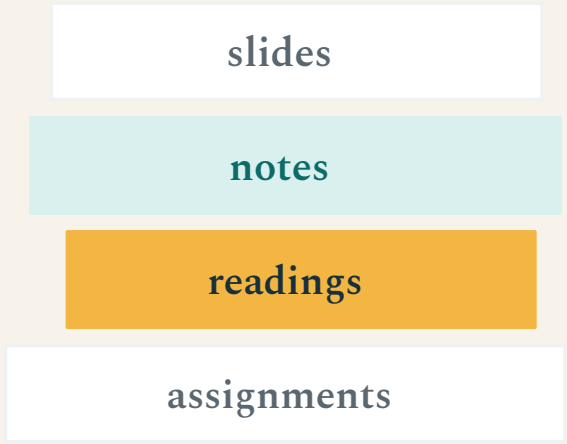
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Natural Language Processing • Fall 2026

The problem students face

Lecture notes can be hard to navigate, even within one module.

- Course materials can be hard to navigate.
- Students may use different wording than the lecture.
- Keyword search can miss relevant explanations.



Vocabulary Gap

Student asks:

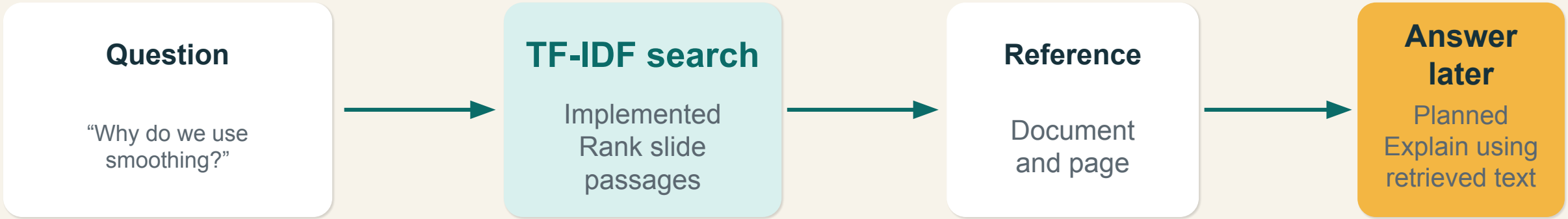
“Why do we use smoothing in language models?”

Course material says:

“handling unseen events and zero probabilities”

CourseLens project and current scope

We search two NLP lectures using natural-language questions. The current prototype returns matching slide text and its source.



Current prototype

Conceptual questions about Language Models and Statistical Task Classification.
Generated answers, broad-query refinement, and follow-up dialogue remain planned.

Background and related work

Published systems already connect student questions with course materials. CourseLens focuses on comparing retrieval methods.

CourseAssist (2024)

- Retrieves course content.
- Classifies intent and breaks down questions.
- Supports CS tutoring.

RetLLM-E (2024)

- Retrieves course documents and prior instructor Q&A.
- Uses that context to answer forum questions.

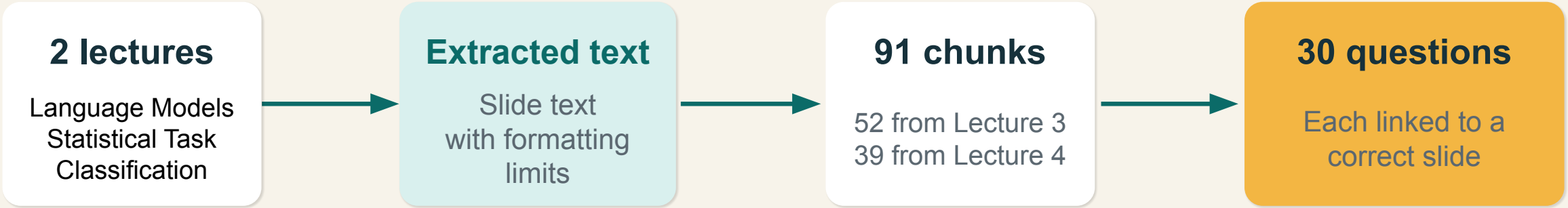
Slides to Chatbots

- Studies university slides and transcripts.
- Compares RAG with continued model training.

Local background: MAGICS Lab AI Tutor, discussed in our first presentation.

Lecture data prepared for retrieval

Two course PDFs provide 91 searchable passages. Each passage represents one slide and keeps its source reference.



Stored fields

chunk_id, text, document, page, n_words

Labeled questions test different wording

The dataset contains 15 direct and 15 paraphrased questions, balanced across the two lectures.

Direct question (Q04)

“What is the Markov assumption in n-gram language models?”

Paraphrased question (Q05)

“Why do n-gram models only look at the previous one or two words instead of everything earlier in the sentence?”

Both questions have the same labeled source

L03_p15 | Language Models, slide 15
“The next token depends only on the last k tokens.”

TF-IDF features and passage ranking

The working baseline represents text with word weights, then compares each question with the lecture passages.

1

Passage text

91 slide chunks

2

Tokenize

Lowercase words

3

TF-IDF

1,235 features

4

Compare

Cosine similarity

5

Rank results

Top passages + source

TF counts word use within a passage. IDF downweights words common across passages.

Learn weights from the lecture text. Reuse them for each question.

The Python prototype

Two small scripts separate searching from measuring retrieval performance.

tfidf_retrieval.py

load_chunks() reads the CSV.
build_index() creates TF-IDF vectors.
search() ranks passages and returns sources.

evaluate_tfidf.py

Runs all 30 labeled questions.
Checks the returned chunk IDs.
Saves predictions and accuracy metrics.

Try a question

python tfidf_retrieval.py "Why do we use smoothing?"

python evaluate_tfidf.py

Top-1 and Top-3 retrieval accuracy

A hit requires the exact chunk ID assigned to the question in the labeled dataset.

Top-1 accuracy

Is the first returned slide the labeled slide?

Top-3 accuracy

Is the labeled slide anywhere in the first three results?

Example ranking	Top-1	Top-3
Correct slide appears first	Hit	Hit
Correct slide appears second	Miss	Hit
Correct slide is absent	Miss	Miss

Top-3 is at least as high as Top-1. We report both on the same questions.

Initial TF-IDF results

The baseline finds the labeled slide more often for direct questions than for paraphrased questions.

Question group	N	Top-1	Top-3
All questions	30	8/30 (26.7%)	16/30 (53.3%)
Direct questions	15	6/15 (40.0%)	12/15 (80.0%)
Paraphrased questions	15	2/15 (13.3%)	4/15 (26.7%)
Lecture 3 questions	15	3/15 (20.0%)	6/15 (40.0%)
Lecture 4 questions	15	5/15 (33.3%)	10/15 (66.7%)

These are overlapping breakdowns of the same 30 questions, not separate datasets.

No parameter search or model comparison has been run yet.

Examples and limits of the baseline

Inspect individual predictions to understand what the aggregate scores hide.

Successful match: Q10

“How does Laplace (add-one) smoothing work?”
Labeled: L03_p32
Retrieved first: L03_p32

Near miss: Q11

“Why does Laplace smoothing overcorrect when the vocabulary is large?”
Labeled: L03_p36
Retrieved: L03_p54, L03_p36, L03_p37

Q10 passes both metrics. Q11 passes Top-3, but misses Top-1.

30 questions make the estimates sensitive to individual examples.
A relevant alternative slide still counts as a miss under exact-ID scoring.
Extracted text can lose image formulas or flatten tables.

Progress and next steps

The dataset and TF-IDF baseline are working. Semantic retrieval and grounded answer generation remain future work.

Milestone 1

Current progress

- 91 slide passages prepared
- 30 questions labeled
- TF-IDF search implemented
- Initial evaluation complete
- Keyword-only baseline pending

Milestone 2

Planned comparison

- Inspect retrieval errors
- Compare word embeddings
- Use the same labeled questions
- Test chunking and ranking
- Reserve questions for final testing

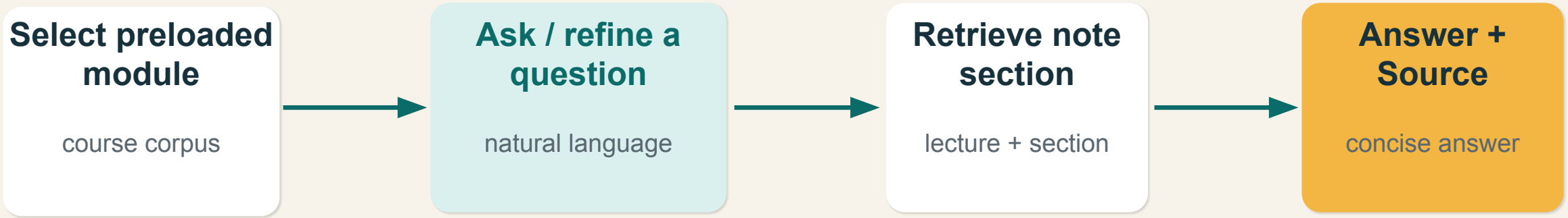
Milestone 3

Planned final system

- Add transformer retrieval
- Generate grounded answers
- Handle unsupported questions
- Add interface and follow-ups
- Compare all methods

Planned final CourseLens experience

The current search prototype is the first step toward a study assistant that explains answers using lecture evidence.



Final goal: a working study assistant and a fair comparison of retrieval methods.

References and project materials

Feng, Liu and Ghosal (2024)

CourseAssist: Pedagogically Appropriate AI Tutor for Computer Science Education

https://arxiv.org/abs/2407.10246

Mitra et al. (2024)

RetLLM-E: Retrieval-Prompt Strategy for Question-Answering on Student Discussion Forums

https://doi.org/10.1609/aaai.v38i21.30368

Dinh, Schumacher and Niehues (2025; revised 2026)

From Slides to Chatbots: Enhancing Large Language Models with University Course Materials

https://arxiv.org/abs/2510.22272

Project repository and baseline implementation

github.com/paria03/CourseLens
TF-IDF documentation: scikit-learn.org/stable/modules/feature_extraction.html

Thank you

Questions and discussion